

SQL Retail Sales Analysis Project

Comprehensive Data Engineering & Exploratory Business Analytics Report

1. Executive Summary

This project showcases a complete end-to-end data pipeline walkthrough using **Structured Query Language (SQL)** to set up databases, engineer target relational schemas, cleanse transactional gaps, and perform deep business exploratory data analysis (EDA). By setting up structured databases and asking targeted strategic questions, this project transforms raw transaction logging datasets into interactive business performance indicators. The methodologies applied isolate consumer demographics, track monthly anomalies, uncover optimal transaction shifts, and highlight high-yield baseline operations.

2. Schema Engineering & Database Architecture

The first structural stage of implementation involves setting up the relational environment under PostgreSQL rules. A baseline table definition ensures strict data types are assigned across variables like financial metrics, calendar dates, timelines, and primary indices.

```
-- Create Database Setup
CREATE DATABASE sql_project_p2;

-- Build Target Table Infrastructure
DROP TABLE IF EXISTS retail_sales;
CREATE TABLE retail_sales (
    transactions_id INT PRIMARY KEY,
    sale_date DATE,
    sale_time TIME,
    customer_id INT,
    gender VARCHAR(15),
    age INT,
    category VARCHAR(15),
    quantity INT,
    price_per_unit FLOAT,
    cogs FLOAT,
    total_sale FLOAT
);
```

3. Aggressive Quality Control & Data Cleansing

To maintain total analytic integrity, records containing missing, incomplete, or corrupted records across critical dimensions are systematically extracted and purged from the operational production database tables.

```
-- Identify and Filter Broken Records
SELECT * FROM retail_sales
WHERE transactions_id IS NULL
    OR sale_date IS NULL
    OR sale_time IS NULL
    OR gender IS NULL
    OR category IS NULL
    OR quantity IS NULL
    OR price_per_unit IS NULL
    OR cogs IS NULL
    OR total_sale IS NULL;

-- Evict Corrupted Integrity Constraints
DELETE FROM retail_sales
WHERE transactions_id IS NULL
    OR sale_date IS NULL
    OR sale_time IS NULL
    OR gender IS NULL
    OR category IS NULL
    OR quantity IS NULL
    OR price_per_unit IS NULL
    OR cogs IS NULL
    OR total_sale IS NULL;
```

4. Strategic Business Analytics Framework

Below are the analytical problem formulations and optimized SQL queries engineered to resolve core business challenges.

Q1. Target Calendar Retrievals

Extract absolute transactional columns for all volume processed on the specific marker of '2022-11-05'.

```
SELECT *
FROM retail_sales
WHERE sale_date = '2022-11-05';
```

Q2. Segmented High-Volume Tracking

Isolate specialized segments ('Clothing') where single-cart item counts reached heavy capacities (≥ 4 items) in November 2022.

```
SELECT *
FROM retail_sales
WHERE category = 'Clothing'
  AND TO_CHAR(sale_date, 'YYYY-MM') = '2022-11'
  AND quantity >= 4;
```

Q3. Net Financial Volume by Category

Aggregate gross total cash inflows alongside sheer volume count partitioned across explicit product verticals.

```
SELECT
  category,
  SUM(total_sale) AS net_sale,
  COUNT(*) AS total_orders
FROM retail_sales
GROUP BY 1;
```

Q4. Consumer Demographics Analysis

Calculate the calculated average buyer age participating in beauty ecosystem product lines.

```
SELECT ROUND(AVG(age), 2) AS avg_age
FROM retail_sales
WHERE category = 'Beauty';
```

Q5. Premium Bracket Threshold Transactions

Filter high-net worth unique tickets scaling past the \$1,000 baseline milestone.

```
SELECT *
FROM retail_sales
WHERE total_sale > 1000;
```

Q6. Cross-Sectional Behavioral Crossings

Enumerate transactional processing patterns mapping gender biases against structural retail inventory category groupings.

```
SELECT
    category,
    gender,
    COUNT(*) AS total_transactions
FROM retail_sales
GROUP BY category, gender
ORDER BY 1;
```

Q7. Dynamic Annual Peak Window Discovery

Establish monthly metrics alongside window performance ranks to find the highest-performing month per year.

```
SELECT year, month, avg_sale
FROM (
    SELECT
        EXTRACT(YEAR FROM sale_date) AS year,
        EXTRACT(MONTH FROM sale_date) AS month,
        AVG(total_sale) AS avg_sale,
        RANK() OVER (
            PARTITION BY EXTRACT(YEAR FROM sale_date)
            ORDER BY AVG(total_sale) DESC
        ) AS rank
    FROM retail_sales
    GROUP BY 1, 2
) as t1
WHERE rank = 1;
```

Q8. VIP Client Lifetime Value (LTV)

Identify the top 5 elite retail spenders contributing highest cumulative funds.

```
SELECT customer_id, SUM(total_sale) AS total_sale
FROM retail_sales
GROUP BY 1
ORDER BY 2 DESC
LIMIT 5;
```

Q9. Unique Market Penetration Metrics

Gauge exact audience scope by distinct individual shoppers touching independent item classification catalogs.

```
SELECT category, COUNT(DISTINCT customer_id) AS unique_cx
FROM retail_sales
GROUP BY category;
```

Q10. Diurnal Operational Velocity Shifts

Condense timestamps via CTE conditional logic down into core daily operational working shifts (Morning, Afternoon, Evening) to count orders.

```
WITH hourly_sale AS (  
    SELECT *,  
        CASE  
            WHEN EXTRACT(HOUR FROM sale_time) < 12 THEN 'Morning'  
            WHEN EXTRACT(HOUR FROM sale_time) BETWEEN 12 AND 17 THEN 'Afternoon'  
            ELSE 'Evening'  
        END AS shift  
    FROM retail_sales  
)  
SELECT shift, COUNT(*) AS total_orders  
FROM hourly_sale  
GROUP BY shift;
```

5. Key Analytical Takeaways

- **Demographic Personalisation:** Targeted segmentation yields deep patterns regarding category trends by gender and age profile clusters.
- **Operational Resource Shifts:** Peak hours (Morning vs Afternoon vs Evening patterns) dictate clear windows for inventory and staff provisioning optimization.
- **Customer Concentrated Yields:** The top 5 customers represent high lifetime value benchmarks, suggesting excellent loyalty loops.